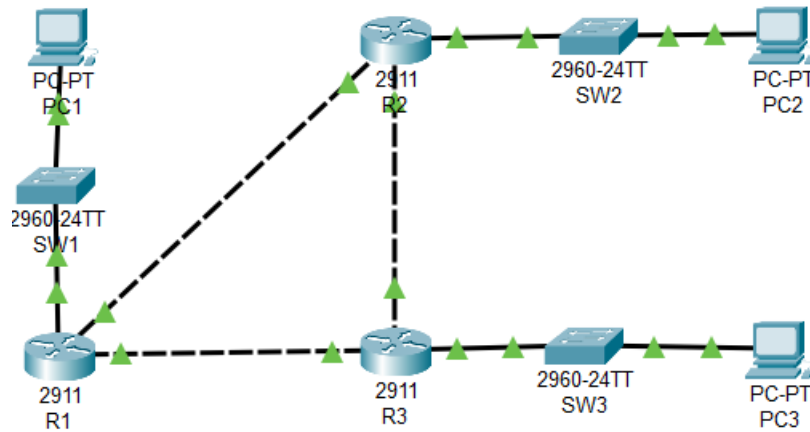




Topology

1. Use CDP (and other commands) to identify and label the missing IP addresses and interface IDs of the devices in the network.

```

PC1

Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::201:63FF:FE8A:B6E0
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.1
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.1.254

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0
  
```

- 192.168.1.1/24 is the IP address of PC1. Since its default gateway is 192.168.1.254, said address would be the IP address of the R1 interface facing SW1.

```

SW1>en
SW1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce    Holdtme    Capability    Platform    Port ID
R1           Gig 0/1          129        R             C2900       Gig 0/2
  
```

- This indicates that SW1's G0/1 interface is connected to R1's G0/2 interface.

```
SW1#show int status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		notconnect	1	auto	auto	10/100BaseTX
Fa0/2		notconnect	1	auto	auto	10/100BaseTX
Fa0/3		notconnect	1	auto	auto	10/100BaseTX
Fa0/4		notconnect	1	auto	auto	10/100BaseTX
Fa0/5		notconnect	1	auto	auto	10/100BaseTX
Fa0/6		notconnect	1	auto	auto	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		connected	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		notconnect	1	auto	auto	10/100BaseTX
Fa0/14		notconnect	1	auto	auto	10/100BaseTX
Fa0/15		notconnect	1	auto	auto	10/100BaseTX
Fa0/16		notconnect	1	auto	auto	10/100BaseTX
Fa0/17		notconnect	1	auto	auto	10/100BaseTX
Fa0/18		notconnect	1	auto	auto	10/100BaseTX
Fa0/19		notconnect	1	auto	auto	10/100BaseTX
Fa0/20		notconnect	1	auto	auto	10/100BaseTX
Fa0/21		notconnect	1	auto	auto	10/100BaseTX
Fa0/22		notconnect	1	auto	auto	10/100BaseTX
Fa0/23		notconnect	1	auto	auto	10/100BaseTX
Fa0/24		notconnect	1	auto	auto	10/100BaseTX
Gig0/1		connected	1	auto	auto	10/100BaseTX
Gig0/2		notconnect	1	auto	auto	10/100BaseTX

- There are two connected interfaces, G0/1 which was already identified as connected to R1, and F0/10, which would naturally be connected to PC1.

```
R1>en
```

```
R1#show cdp neighbors
```

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
R2	Gig 0/1	146	R	C2900	Gig 0/0
SW1	Gig 0/2	146	S	2960	Gig 0/1
R3	Gig 0/0	146	R	C2900	Gig 0/1

- This says that R1's G0/1 is connected to R2's G0/0, R1's G0/2 is connected to SW1's G0/1, and R1's G0/0 is connected to R3's G0/1.

```
R1#show int g0/0
```

```
GigabitEthernet0/0 is up, line protocol is up (connected)  
Hardware is CN Gigabit Ethernet, address is 0010.119d.2501 (bia 0010.119d.2501)  
Internet address is 10.0.13.1/30
```

```
R1#show int g0/1
```

```
GigabitEthernet0/1 is up, line protocol is up (connected)  
Hardware is CN Gigabit Ethernet, address is 0010.119d.2502 (bia 0010.119d.2502)  
Internet address is 10.0.12.1/30
```

- The following commands indicate that the IP address of R1's g0/0 and g0/1 interfaces are 10.0.13.1/30 and 10.0.12.1/30.

```

R2>en
R2#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID      Local Intrfce  Holdtme  Capability  Platform  Port ID
R1             Gig 0/0      172      R           C2900     Gig 0/1
SW2            Gig 0/1      172      S           2960      Gig 0/2
R3             Gig 0/2      172      R           C2900     Gig 0/2

```

- This indicates that R2's G0/0 is connected to R1's G0/1, R2's G0/1 is connected to SW2's G0/2, and R2's G0/2 is connected to R3's G0/2.

```

R2#show int g0/0
GigabitEthernet0/0 is up, line protocol is up (connected)
  Hardware is CN Gigabit Ethernet, address is 00e0.8f6c.6101 (bia 00e0.8f6c.6101)
  Internet address is 10.0.12.2/30

```

```

R2#show int g0/1
GigabitEthernet0/1 is up, line protocol is up (connected)
  Hardware is CN Gigabit Ethernet, address is 00e0.8f6c.6102 (bia 00e0.8f6c.6102)
  Internet address is 192.168.2.254/24

```

```

R2#show int g0/2
GigabitEthernet0/2 is up, line protocol is up (connected)
  Hardware is CN Gigabit Ethernet, address is 00e0.8f6c.6103 (bia 00e0.8f6c.6103)
  Internet address is 10.0.23.1/30

```

- The following commands indicate that the IP address of R2's g0/0, g0/1 and g0/2 interfaces are 10.0.12.2/30, 192.168.2.254/24 and 10.0.23.1/30, respectively.

```

R3>en
R3#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID      Local Intrfce  Holdtme  Capability  Platform  Port ID
R1             Gig 0/1      173      R           C2900     Gig 0/0
R2             Gig 0/2      173      R           C2900     Gig 0/2
SW3            Gig 0/0      173      S           2960      Gig 0/1

```

- This indicates that R3's G0/1 is connected to R1's G0/0, R3's G0/2 is connected to R2's G0/2, and R3's G0/0 is connected to SW3's G0/1.

```

R3#show int g0/1
GigabitEthernet0/1 is up, line protocol is up (connected)
  Hardware is CN Gigabit Ethernet, address is 0002.1692.3c02 (bia 0002.1692.3c02)
  Internet address is 10.0.13.2/30

```

```

R3#show int g0/0
GigabitEthernet0/0 is up, line protocol is up (connected)
  Hardware is CN Gigabit Ethernet, address is 0002.1692.3c01 (bia 0002.1692.3c01)
  Internet address is 192.168.3.254/24

```

```

R3#show int g0/2
GigabitEthernet0/2 is up, line protocol is up (connected)
  Hardware is CN Gigabit Ethernet, address is 0002.1692.3c03 (bia 0002.1692.3c03)
  Internet address is 10.0.23.2/30

```

- The following commands indicate that the IP address of R3's g0/0, g0/1 and g0/2 interfaces are 192.168.3.254/24, 10.0.13.2/30, and 10.0.23.2/30 respectively.

```
SW2>en
SW2#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID      Local Intrfce  Holdtme  Capability  Platform  Port ID
R2              Gig 0/2        122      R           C2900     Gig 0/1
```

- This indicates that SW2's G0/2 interface is connected to R2's G0/1 interface.

```
SW2#show int status
Port      Name      Status      Vlan      Duplex  Speed  Type
Fa0/1     Fa0/1     connected   1         auto    auto   10/100BaseTX
Fa0/2     Fa0/2     notconnect  1         auto    auto   10/100BaseTX
Fa0/3     Fa0/3     notconnect  1         auto    auto   10/100BaseTX
Fa0/4     Fa0/4     notconnect  1         auto    auto   10/100BaseTX
Fa0/5     Fa0/5     notconnect  1         auto    auto   10/100BaseTX
Fa0/6     Fa0/6     notconnect  1         auto    auto   10/100BaseTX
Fa0/7     Fa0/7     notconnect  1         auto    auto   10/100BaseTX
Fa0/8     Fa0/8     notconnect  1         auto    auto   10/100BaseTX
Fa0/9     Fa0/9     notconnect  1         auto    auto   10/100BaseTX
Fa0/10    Fa0/10    notconnect  1         auto    auto   10/100BaseTX
Fa0/11    Fa0/11    notconnect  1         auto    auto   10/100BaseTX
Fa0/12    Fa0/12    notconnect  1         auto    auto   10/100BaseTX
Fa0/13    Fa0/13    notconnect  1         auto    auto   10/100BaseTX
Fa0/14    Fa0/14    notconnect  1         auto    auto   10/100BaseTX
Fa0/15    Fa0/15    notconnect  1         auto    auto   10/100BaseTX
Fa0/16    Fa0/16    notconnect  1         auto    auto   10/100BaseTX
Fa0/17    Fa0/17    notconnect  1         auto    auto   10/100BaseTX
Fa0/18    Fa0/18    notconnect  1         auto    auto   10/100BaseTX
Fa0/19    Fa0/19    notconnect  1         auto    auto   10/100BaseTX
Fa0/20    Fa0/20    notconnect  1         auto    auto   10/100BaseTX
Fa0/21    Fa0/21    notconnect  1         auto    auto   10/100BaseTX
Fa0/22    Fa0/22    notconnect  1         auto    auto   10/100BaseTX
Fa0/23    Fa0/23    notconnect  1         auto    auto   10/100BaseTX
Fa0/24    Fa0/24    notconnect  1         auto    auto   10/100BaseTX
Gig0/1    Gig0/1    notconnect  1         auto    auto   10/100BaseTX
Gig0/2    Gig0/2    connected   1         auto    auto   10/100BaseTX
```

- There are two connected interfaces, G0/2 which was already identified as connected to R2, and F0/1, which would naturally be connected to PC2.

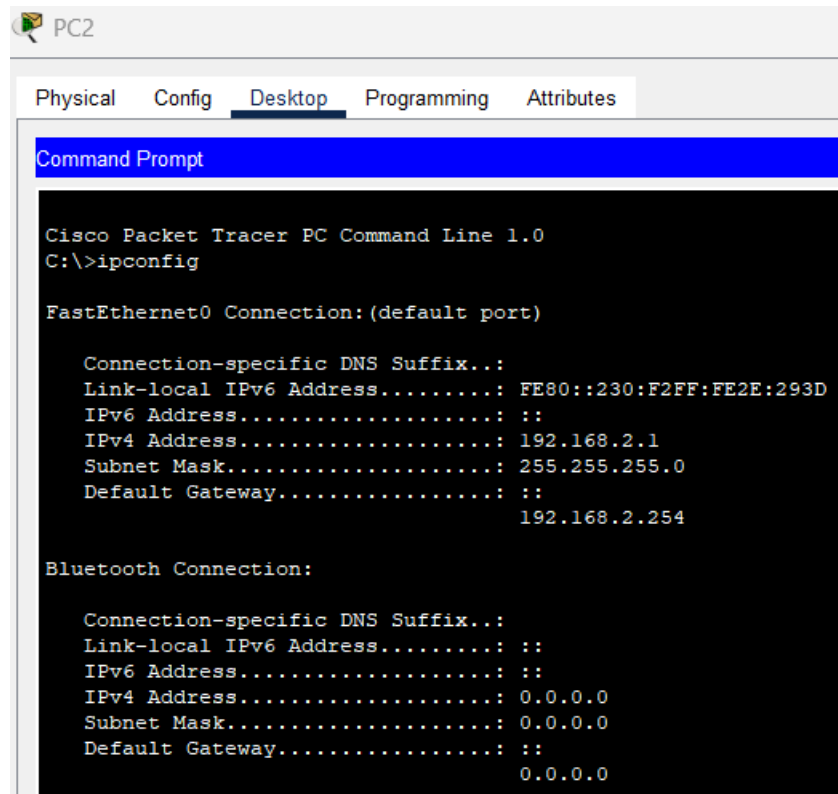
```
SW3>en
SW3#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID      Local Intrfce  Holdtme  Capability  Platform  Port ID
R3              Gig 0/1        123      R           C2900     Gig 0/0
```

- This indicates that SW3's G0/1 interface is connected to R3's G0/0 interface.

SW3#show int status

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		notconnect	1	auto	auto	10/100BaseTX
Fa0/2		notconnect	1	auto	auto	10/100BaseTX
Fa0/3		notconnect	1	auto	auto	10/100BaseTX
Fa0/4		notconnect	1	auto	auto	10/100BaseTX
Fa0/5		notconnect	1	auto	auto	10/100BaseTX
Fa0/6		notconnect	1	auto	auto	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		notconnect	1	auto	auto	10/100BaseTX
Fa0/14		notconnect	1	auto	auto	10/100BaseTX
Fa0/15		notconnect	1	auto	auto	10/100BaseTX
Fa0/16		notconnect	1	auto	auto	10/100BaseTX
Fa0/17		notconnect	1	auto	auto	10/100BaseTX
Fa0/18		notconnect	1	auto	auto	10/100BaseTX
Fa0/19		notconnect	1	auto	auto	10/100BaseTX
Fa0/20		notconnect	1	auto	auto	10/100BaseTX
Fa0/21		notconnect	1	auto	auto	10/100BaseTX
Fa0/22		notconnect	1	auto	auto	10/100BaseTX
Fa0/23		notconnect	1	auto	auto	10/100BaseTX
Fa0/24		connected	1	auto	auto	10/100BaseTX
Gig0/1		connected	1	auto	auto	10/100BaseTX
Gig0/2		notconnect	1	auto	auto	10/100BaseTX

- There are two connected interfaces, G0/1 which was already identified as connected to R3, and F0/24, which would naturally be connected to PC3.



- 192.168.2.1/24 is the IP address of PC2. Since its default gateway is 192.168.2.254, said address would be the IP address of the R2 interface facing SW2.

PC3

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

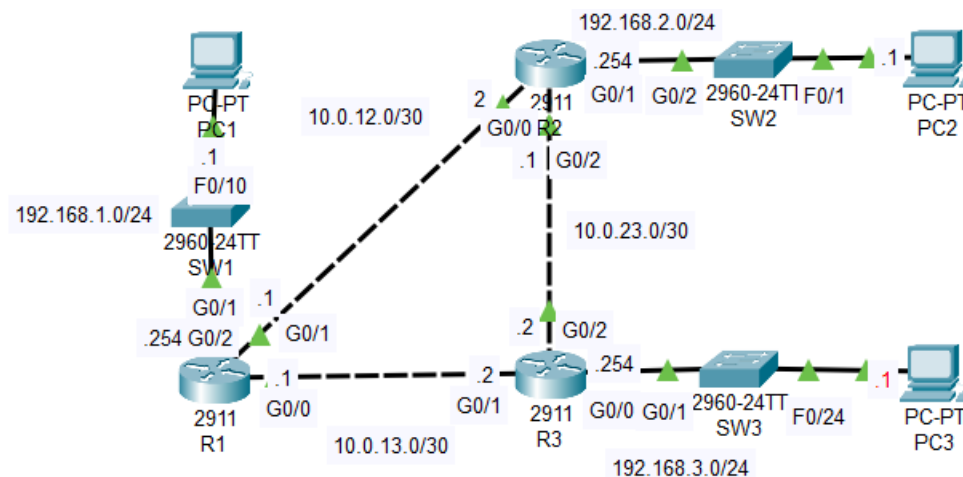
FastEthernet0 Connection:(default port)

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: FE80::240:BFF:FE3A:CA62
IPv6 Address.....: ::
IPv4 Address.....: 192.168.3.1
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                        192.168.3.254

Bluetooth Connection:

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
                        0.0.0.0
```

- 192.168.3.1/24 is the IP address of PC3. Since its default gateway is 192.168.3.254, said address would be the IP address of the R3 interface facing SW3.



2. Disable CDP on the switch interfaces currently connected to PCs.

```

SW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#int f0/10
SW1(config-if)#do show cdp int f0/10
FastEthernet0/10 is up, line protocol is up
  Sending CDP packets every 60 seconds
  Holdtime is 180 seconds
SW1(config-if)#no cdp enable
SW2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW2(config)#int f0/1
SW2(config-if)#do show cdp int f0/1
FastEthernet0/1 is up, line protocol is up
  Sending CDP packets every 60 seconds
  Holdtime is 180 seconds
SW2(config-if)#no cdp enable
SW3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW3(config)#int f0/24
SW3(config-if)#do show cdp int f0/24
FastEthernet0/24 is up, line protocol is up
  Sending CDP packets every 60 seconds
  Holdtime is 180 seconds
SW3(config-if)#no cdp enable

```

3. Disable CDP globally on each network device.
4. Enable LLDP globally on each network device, and enable Tx/Rx on the interfaces connected to other network devices. *Tx/Rx are currently disabled on all interfaces

```

R1(config)#no cdp run
R1(config)#lldp run
R1(config)#int range g0/0 - 2
R1(config-if-range)#lldp transmit
R1(config-if-range)#lldp receive
R2(config)#no cdp run
R2(config)#lldp run
R2(config)#int range g0/0 - 2
R2(config-if-range)#lldp transmit
R2(config-if-range)#lldp receive
R3(config)#no cdp run
R3(config)#lldp run
R3(config)#int range g0/0 - 2
R3(config-if-range)#lldp transmit
R3(config-if-range)#lldp receive
SW1(config)#no cdp run
SW1(config)#lldp run
SW1(config)#int g0/1
SW1(config-if)#lldp transmit
SW1(config-if)#lldp receive
SW2(config)#no cdp run
SW2(config)#lldp run
SW2(config)#int g0/2
SW2(config-if)#lldp transmit
SW2(config-if)#lldp receive
SW3(config)#no cdp run
SW3(config)#lldp run
SW3(config)#int g0/1
SW3(config-if)#lldp transmit
SW3(config-if)#lldp receive

```